

Decrypt an Encrypted Message

Introduction

In this lab, I practiced using Linux commands to decrypt hidden and encrypted messages. I worked with a Caesar cipher and then used OpenSSL to decrypt an encrypted file.

Scenario

All files in the `/home/analyst` directory were encrypted. I needed to find the instructions, solve the Caesar cipher, and use the recovered command to decrypt the main file.

Task 1: Read the File

I first listed the files in the home directory:

```
ls /home/analyst
```

The directory contained:

```
Q1.encrypted  README.txt  caesar
```

I then read the README file:

```
cat README.txt
```

The message instructed me to look for a hidden file inside the `caesar` directory.

Task 2: Find and Decrypt the Hidden File

I moved to the `caesar` directory:

```
cd caesar
```

I used `ls -a` to show hidden files:

```
ls -a
```

The hidden file was:

```
.leftShift3
```

I read the file:

```
cat .leftShift3
```

The message was encrypted using a Caesar cipher with a shift of three letters.

I decrypted it using:

```
cat .leftShift3 | tr "d-za-cD-ZA-C" "a-zA-Z"
```

This revealed the OpenSSL command needed to decrypt the main file:

```
openssl aes-256-cbc -pbkdf2 -a -d -in Q1.encrypted -out Q1.recovered -k  
ettubrute
```

I then returned to my home directory:

```
cd ~
```

Task 3: Decrypt the Encrypted File

I used the recovered OpenSSL command:

```
openssl aes-256-cbc -pbkdf2 -a -d -in Q1.encrypted -out Q1.recovered -k  
ettubrute
```

I checked the files:

```
ls
```

A new file named `Q1.recovered` was created.

I read the recovered file:

```
cat Q1.recovered
```

The message confirmed that the file had been successfully decrypted.

Commands I Practiced

- `ls`
- `cat`
- `cd`
- `ls -a`
- `tr`
- `openssl`

What I Learned

I learned how to find hidden files in Linux and how a Caesar cipher can be reversed using the `tr` command. I also practiced using OpenSSL to decrypt an AES-256-CBC encrypted file.

Cybersecurity Connection

Encryption protects data from unauthorized access. Security analysts need to understand both encryption and decryption so they can work with protected data and understand how cryptographic controls are used.

Demonstration

Screenshot of the completed decryption tasks:

```
analyst@409198b36688:~$ ls /home/analyst
Q1.encrypted  README.txt  caesar
analyst@409198b36688:~$ cat README.txt
Hello,
All of your data has been encrypted. To recover your data, you will need to solve a cipher. To get started look for a hidden file in the caesar subdirectory.
analyst@409198b36688:~$ cd caesar
analyst@409198b36688:~/caesar$ ls -a
.  ..  .leftShift3
analyst@409198b36688:~/caesar$ cat .leftShift3
Lq rughu wr uhfryhu brxu ilohv brx zloo qhhg wr hqwhu wkh iroorzlqj frppdgg:

rshqvvo dhv-256-fef -sengi2 -d -g -lq Tl.hqfubswhg -rxw Tl.uhfryhuhg -n hwwxeuxw
n
analyst@409198b36688:~/caesar$ cat .leftShift3 | tr "d-za-cD-ZA-C" "a-zA-Z"
In order to recover your files you will need to enter the following command:

openssl aes-256-cbc -pbkdf2 -a -d -in Q1.encrypted -out Q1.recovered -k ettubrut
e
analyst@409198b36688:~/caesar$
```

```
analyst@409198b36688:~/caesar$ cd ~
analyst@409198b36688:~$ openssl aes-256-cbc -pbkdf2 -a -d -in Q1.encrypted -out
Q1.recovered -k ettubrut
analyst@409198b36688:~$ ls
Q1.encrypted  Q1.recovered  README.txt  caesar
analyst@409198b36688:~$ cat Q1.recovered
If you are able to read this, then you have successfully decrypted the classic c
ipher text. You recovered the encryption key that was used to encrypt this file.
Great work!
analyst@409198b36688:~$ █
```

Conclusion

In this lab, I found a hidden file, decrypted a Caesar cipher, and used the recovered key and OpenSSL command to decrypt an encrypted file successfully.